

Using Auto-Scaling Group

EC2 > Launch templates > Create launch template

Success
Successfully created Server101(lt-0008be67cb69848c8).

Actions log

Next Steps

Launch an instance

With On-Demand Instances, you pay for compute capacity by the second (for [Create launch template](#) seconds) or by the hour (for all other operating systems) with no long-term commitments or upfront payments. Launch an On-Demand Instance from your launch template.
[Launch instance from this template](#)

Create an Auto Scaling group from your template

Amazon EC2 Auto Scaling helps you maintain application availability and allows you to scale your Amazon EC2 capacity up or down automatically according to conditions you define. You can use Auto Scaling to help ensure that you are running your desired number of Amazon EC2 instances during demand spikes to maintain performance and decrease capacity during lulls to reduce costs.
[Create Auto Scaling group](#)

Create Spot Fleet

A Spot Instance is an unused EC2 instance that is available for less than the On-Demand price. Because Spot Instances enable you to request unused EC2 instances at steep discounts, you can lower your Amazon EC2 costs significantly. The hourly price for a Spot Instance (of each instance type in each Availability Zone) is set by Amazon EC2, and adjusted gradually based on the long-term supply of and demand for Spot Instances. Spot instances are well-suited for data-analysis, batch jobs, background processing, and optional tasks.
[Create Spot Fleet](#)

View launch templates

Name

Auto Scaling group name

Enter a name to identify the group.

ASG-1

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

Server101

[Create a launch template](#)

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-03ef38521848dce89 (Lab6-VPC)
172.31.0.0/16 Default

[Create a VPC](#)

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

usw1-az1 (us-west-1a) | subnet-0be9eafa2a68ebb8e (Public-Subnet-1a)
172.31.16.0/20 Default

usw1-az3 (us-west-1c) | subnet-0c5fa4dd8ca549962 (Public-Subnet-1b)
172.31.0.0/20 Default

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

Balanced best effort

If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

Balanced only

If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

2

↕

Equal or less than desired capacity

Max desired capacity

3

↕

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☐ Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Instance maintenance policy [Info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

Mixed behavior

☒ No policy

For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.

Prioritize availability

☐ Launch before terminating

Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.

Control costs

☐ Terminate and launch

Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.

Flexible

☐ Custom behavior

Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.

ASG-1

Auto Scaling group details

ASG-1 Capacity overview

Edit

`arn:aws:autoscaling:us-west-1:204312534875:autoScalingGroup:e134f5d9-9c3b-4349-9e6d-1c35d2ef271e:autoScalingGroupName/ASG-1`

Desired capacity

2

Scaling limits

2 - 3

Desired capacity type

Units (number of instances)

Status

Updating capacity

Date created

Fri Mar 06 2026 21:41:59 GMT-0800 (Pacific Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Launch template

Edit

Launch template

`lt-0008be67cb69848c8`
Server101

Version

Default

Description

WebServer

[View details in the launch template console](#)

AMI ID

`ami-038f4d4c8824bfed9`

Security groups

-

Storage (volumes)

-

Instance type

-

Security group IDs

`sg-03a7b834069987598`

Key pair name

lab6-key

Owner

`arn:aws:iam::204312534875:root`

Create time

Fri Mar 06 2026 21:20:44 GMT-0800 (Pacific Standard Time)

Request Spot Instances

No

Network

Edit

Availability Zones

`usw1-az1` (us-west-1a)
`usw1-az3` (us-west-1c)

Subnet ID

`subnet-Obe9eafa2a68ebb8e`
 `subnet-0c5fa4dd8ca549962`

Availability Zone distribution

Balanced best effort

☐	<code>i-0f41c782a75a1a788</code>	Running	<code>t3.micro</code>	3/3 checks passed View alarms +	<code>us-west-1a</code>	<code>ec2-50-18-76-89.us-we...</code>	50.18.76.89
☐	<code>i-0130eadc609da2476</code>	Shutting-d...	<code>t3.micro</code>	- View alarms +	<code>us-west-1c</code>	<code>ec2-52-53-198-144.us-...</code>	52.53.198.144

Activity history (4)

Filter activity history

Filter by a date and time range

< 1 >

Status	Description	Cause	Start time	End time
Successful	Launching a new EC2 instance: i-0412e8da632dcd36a	At 2026-03-07T05:46:08Z an instance was launched in response to an unhealthy instance needing to be replaced.	2026 March 06, 09:46:10 PM -08:00	2026 March 06, 09:46:15 PM -08:00
Successful	Terminating EC2 instance: i-0130eadc609da2476	At 2026-03-07T05:46:07Z an instance was taken out of service in response to an EC2 health check indicating it has been terminated or stopped.	2026 March 06, 09:46:08 PM -08:00	2026 March 06, 09:46:29 PM -08:00
Successful	Launching a new EC2 instance: i-0130eadc609da2476	At 2026-03-07T05:41:59Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-03-07T05:42:01Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 March 06, 09:42:03 PM -08:00	2026 March 06, 09:42:11 PM -08:00
Successful	Launching a new EC2 instance: i-0f41c782a75a1a788	At 2026-03-07T05:41:59Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2026-03-07T05:42:01Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2026 March 06, 09:42:03 PM -08:00	2026 March 06, 09:42:08 PM -08:00

Elastic Load Balancing

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB

Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB

IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB

Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

TG-01

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 5/32

Protocol

Protocol for communication between the load balancer and targets.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in

Successfully created the target group: TG-01. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab.

TG-01

Actions

Details

arn:aws:elasticloadbalancing:us-west-1:204312534875:targetgroup/TG-01/bad107a723e5060e

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

vpc-03ef38521848dce89

IP address type

IPv4

Load balancer

None associated

0

Total targets

0

Healthy

0

Unhealthy

0

Unused

0

Initial

0

Draining

0

Anomalous

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (0)

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

< 1 >

Instance ID

Name

Port

Zone

Health status

Health status details

Administrative o...

Override details

Launch time

Create Load Balancer

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

LB-01

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Scheme can't be changed after the load balancer is created.

Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

IPv4

Includes only IPv4 addresses.

Dualstack

Includes IPv4 and IPv6 addresses.

Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups.

vpc-03ef38521848dce89 (Lab6-VPC)
172.31.0.0/16

(default)

Create VPC

IP pools

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the Amazon VPC IP Address Manager console.

Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

us-west-1a (usw1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0be9eafa2a68ebb8e (Public-Subnet-1a)
172.31.16.0/20

us-west-1c (usw1-az3)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0c5fa4dd8ca549962 (Public-Subnet-1b)
172.31.0.0/20

🟢 **Successfully created load balancer: LB-01**
It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

📘 **Introducing ALB target optimizer**
Target optimizer lets you enforce a maximum number of requests per target using an ALB-provided agent, improving success rates, latency, and efficiency. [Learn more](#)

LB-01 🔄 Actions

▼ Details

Load balancer type
Application

Scheme
Internet-facing

Status
🔄 Provisioning

Hosted zone
Z368ELLRRE2KJ0

VPC
[vpc-03ef38521848dce89](#)

Availability Zones
[subnet-0be9eafa2a68ebb8e](#) us-west-1a (usw1-az1)
[subnet-0c5fa4dd8ca549962](#) us-west-1c (usw1-az3)

Load balancer IP address type
IPv4

Date created
March 6, 2026, 22:00 (UTC-08:00)

Load balancer ARN
[arn:aws:elasticloadbalancing:us-west-1:204312534875:loadbalancer/app/LB-01/327c8d20abc039d7](#)

DNS name [Info](#)
[LB-01-1800894734.us-west-1.elb.amazonaws.com](#) (A Record)

[Listeners and rules](#) | [Network mapping](#) | [Resource map](#) | [Security](#) | [Monitoring](#) | [Integrations](#) | [Attributes](#) | [Capacity](#) | [Tags](#)

Listeners and rules (1) [Info](#) 🔄 Manage rules Manage listener Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

☐ Protocol:Port

☐ Default action

☐ Rules

☐ ARN

☐ Security policy

☐ Default SSL/TLS cer...

☐ mTLS

☐ Trust st

☐ HTTP:80	• Forward to target group TG-01: 1 (100%) Target group stickiness: Off	1 rule	A...	Not applicable	Not applicable	Not applicable	Not app
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Adding Instances

Load balancing - optional

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Load balancers

☒ **Application, Network or Gateway Load Balancer target groups**
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

TG-01 | HTTP

Application Load Balancer: LB-01

☐ **Classic Load Balancers**

Create and attach new load balancers

[Add a new load balancer](#)

Registered targets (2) [Info](#) 🔗 Anomaly mitigation: Not applicable 🔄 Deregister Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

☐ Instance ID

☐ Name

☐ Port

☐ Zone

☐ Health status

☐ Health status details

☐ Administrative o...

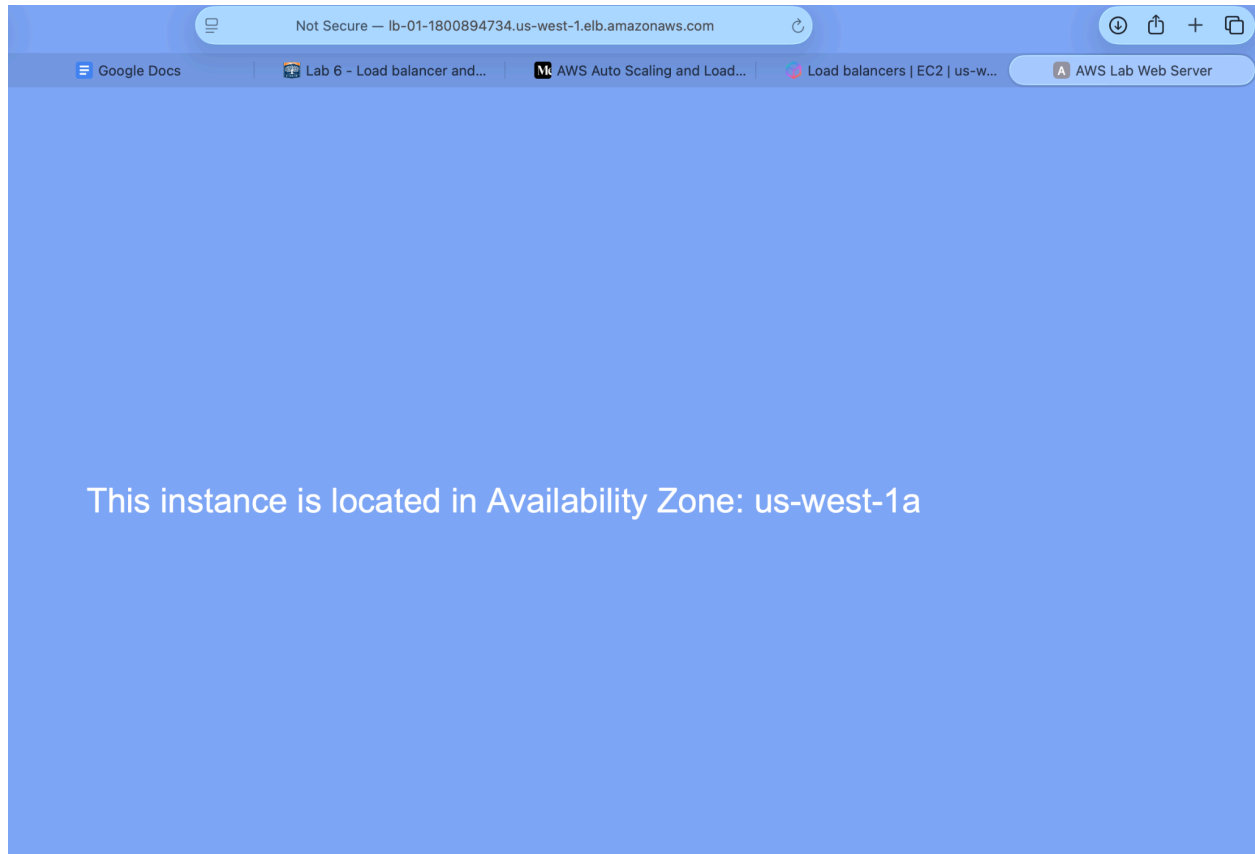
☐ Override details

☐ Launch...

☐ Anomaly

☐ i-0412e8d9632dcd36a	-	80	us-west-1c (us...	🟢 Healthy	-	🔄 No override	No override is curre...	March 6, ...	🟢 Norma
☐ i-0f41c782a75a1a788	-	80	us-west-1a (us...	🟢 Healthy	-	🔄 No override	No override is curre...	March 6, ...	🟢 Norma

Test Load Balancers



Create Scaling Policy

Group size

Specify the size of the Auto Scaling group by changing the desired capacity. You can also specify the maximum scaling limits.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

2

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

2

Equal or less than desired capacity

Max desired capacity

4

Equal or greater than desired capacity

Cancel

Update

Network

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The d

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets



- us-west-1a | subnet-0be9eafa2a68ebb8e (Public-Subnet-1a) ✕
172.31.16.0/20 Default
- us-west-1a | subnet-041e9ea9afe74fc62 (Public-Subnet-1d) ✕
172.31.32.0/20
- us-west-1c | subnet-03875fdb573e279b (Public-Subnet-1c) ✕
172.31.80.0/20
- us-west-1c | subnet-0c5fa4dd8ca549962 (Public-Subnet-1b) ✕
172.31.0.0/20 Default

Create a subnet

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

us-west-1 region seems to only have 2 available zones?

Target Tracking Policy

Policy type

Target tracking scaling

Enabled or disabled

Enabled

Execute policy when

As required to maintain Application Load Balancer request count per target at 50

Take the action

Add or remove capacity units as required

Instances need

300 seconds to warm up before including in metric

Scale in

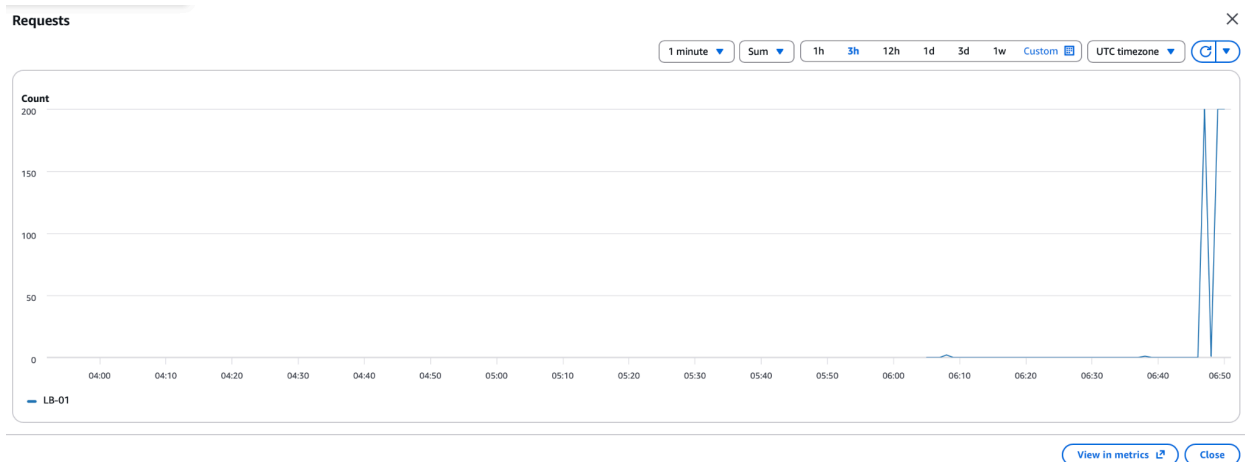
Enabled

Alarms (2)

Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
TargetTracking-ASG-1-AlarmLow-ec890a43-9b53-4f48-bc4a-5dece6be14ec	In alarm	Actions enabled	-	2026-03-07 06:42:56	RequestCountPerTarget < 35 for 15 datapoints within 15 minutes
TargetTracking-ASG-1-AlarmHigh-25ad9bc4-22fc-4559-88f9-dda2a50080fb	OK	Actions enabled	-	2026-03-07 06:42:25	RequestCountPerTarget > 50 for 3 datapoints within 3 minutes

Load balancer: LB-01





Cleaning Up Resources

Auto Scaling groups (1) [Info](#)

Last updated less than a minute ago [Refresh](#) [Launch configurations](#) [Launch templates](#) [Actions](#) [Create Auto Scaling group](#)

☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availability Zones	Creation time
☐	ASG-1	Server101 Version Default	2	Deleting	0	0	0	2 Availability Zones	Fri Mar 06 2026 21:41:...

Successfully deleted load balancer: arn:aws:elasticloadbalancing:us-west-1:204312534875:loadbalancer/app/LB-01/327c8d20abc039d7.

Load balancers [What's new?](#) [Refresh](#) [Actions](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

☐	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security
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